

```
1  #include <stdio.h>
2  void push();
3  void pop();
4  void peek();
5  int N=5;
6  int stack[5];
7  int top=-1;
8  int main()
9  {
10     int ch;
11
12     do
13     {printf("Enter your choice:1(push)/2(pop)/3(peek)/4(exit):\n");
14      scanf("%d",&ch);
15      switch(ch)
16      {
17      case 1 :
18          push();
19          break;
20      case 2 :
21          pop();
22          break;
23      case 3 :
24          peek();
25          break;
26      case 4:
27          break;
28      default :
29          printf("choice is out of range");
30          break;
31      }
32     }while(ch!=4);
33     return 0;
34 }
35 void push()
36 {
37     int x;
38     printf("enter data:");
39     scanf("%d",&x);
40     if(top==N-1)
```

```
39     scanf("%d",&x);
40     if(top==N-1)
41     {
42         printf("overflow..cannot enter data\n");
43     }
44     else
45     {
46         top++;
47         stack[top]=x;
48     }
49 }
50 void pop()
51 {
52     int item;
53     if(top== -1)
54     {
55         printf("underflow...stack is empty\n");
56     }
57     else
58     {
59         item=stack[top];
60         top--;
61         printf("%d",item);
62         printf("\n");
63     }
64 }
65 void peek()
66 {
67     if(top== -1)
68     {
69         printf("underflow...\n");
70     }
71     else
72     {
73         printf("%d\n",stack[top]);
74         printf("\n");
75     }
76 }
77
```

C:\Users\student\Desktop\1BI X + v

```
Enter your choice:1(push)/2(pop)/3(peek)/4(exit):
2
underflow...stack is empty
Enter your choice:1(push)/2(pop)/3(peek)/4(exit):
3
underflow...
Enter your choice:1(push)/2(pop)/3(peek)/4(exit):
1
enter data:1
Enter your choice:1(push)/2(pop)/3(peek)/4(exit):
3
1

Enter your choice:1(push)/2(pop)/3(peek)/4(exit):
2
1
Enter your choice:1(push)/2(pop)/3(peek)/4(exit):
4

Process returned 0 (0x0)   execution time : 222.436 s
Press any key to continue.
|
```